COURSE SYLLABUS IN

COMPUTER PROGRAMMING 1

Academic Year 2023 to 2024

Course Code	**CCP 1101**				
Course Title	**COMPUTER PROGRAMMING 1 (PYTHON)**				
Credit Units	**Lecture**	**2 units**			
	Laboratory / Studio	**1 unit**			
Course Type	☐ **PURE ONSITE [Face-to-Face]**	☐ **PURE OFFSITE [Online Distance Learning]**	☑ **HYBRID [Onsite + Offsite]**	☐ **Others. Please specify.**	
Pre-requisite(s)	**Course Code**				
	Course Title				

Course Description:

Procedural programming, algorithm design, and language constructs to high level programming language.

COURSE OUTCOMES

Upon completion of the course, the learner will be able to:

1. Implement a given algorithm as a computer program in Python
2. Adapt and combine standard algorithms to solve a given problem (includes numerical as well as non-numerical algorithms)
3. Adequately use standard programming constructs: repetition, selection, functions, composition, modules, aggregated data (such as arrays and lists)
4. Explain what a given Python program does
5. Identify and repair coding errors in a program

LEARNING MATRIX									
WEEK	**HOUR**		**LEARNING OUTCOME**	**CONTENT**	**TEACHING-LEARNING ACTIVITY**				**ASSESSMENT**
					Title	**Modality** **O**-Onsite **A**-Offsite Asynchronous **S**-Offsite Synchronous			**Resource**	
	Lec	Lab				**O**	**A**	**S**		
1st - 2nd	4	6	At the end of the lesson, the learner will be able to: • Discuss classroom policy (Class Performance, Attendance Monitoring, and Grading System) • Use flowcharts and pseudocodes to represent program modules • Compose and run a simple Python program that accepts input and displays output	**1. Orientation and overview of the course** **2. Introduction** • Overview of computing • Flowcharting/ Pseudocode • Environment and Compiler • Program Structure • I/O statements	• **Flowcharting activity**	✓			• PowerPoint presentation	• Seatwork • Quiz • Assignment • Class Participation
					• **Multimedia Presentation** • **Lecture** • **Code Demonstration** • **Research**	✓				
3rd - 6th	8	12	• Initialize and use variables with appropriate names • Use the appropriate data type • Perform string formatting operations • Construct arithmetic expressions • Construct a simple Python program that performs inputs, calculations and outputs	**3. Values and Variables** • Identifiers • Variables and Assignment • Integer, String, Floating-Point • String Control Codes **4. Operators and Expressions** • Expressions • Operators	• **Programming Activity**	✓	✓		• PowerPoint presentation	• Seatwork • Quiz • Assignment • Class Participation • Major Exam
					• **Multimedia Presentation** • **Lecture** • **Code Demonstration**	✓				
			PRELIMINARY EXAMINATION							

LEARNING MATRIX

WEEK	HOUR		LEARNING OUTCOME	CONTENT	TEACHING-LEARNING ACTIVITY					ASSESSMENT
					Title	Modality **O**-Onsite, **A**-Offsite Asynchronous, **S**-Offsite Synchronous			Resource	
	Lec	Lab				**O**	**A**	**S**		
7th - 9th	6	9	At the end of the lesson, the learner will be able to: • Write a loop to repeat a sequence of actions a fixed number of times • Write a loop that counts down and a loop that counts up • Write an entry-controlled loop that halts when a condition becomes false • Construct appropriate conditions for condition-controlled loops and selection statements	**5. Loops** • While statement • For statement • While/Else and For/Else	• **Programming Activity**	✓	✓		• PowerPoint presentation	• Seatwork • Quiz • Assignment • Class Participation
					• **Multimedia Presentation** • **Lecture** • **Code Demonstration**	✓				
10th – 11th	4	6	• Explain why functions are useful in structuring code in a program • Employ top-down design to assign tasks to functions • Define a function with required and optional parameters	**6. Functions** • Functions and Modules • Built-in Functions • Standard Mathematical Functions • Parameter Passing	• **Programming Activity**	✓	✓		• PowerPoint presentation	• Seatwork • Quiz • Assignment • Class Participation
					• **Multimedia Presentation** • **Lecture** • **Code Demonstration**	✓				
12th	2	3	• Construct lists and access items in those lists • Use methods to manipulate lists • Perform traversals of lists to process items in the lists • Define simple functions that expect parameters and return	**7. Lists** • List and List Traversal • Building Lists • List Membership • Line Assignment and Equivalence • List Bounds • Slicing	• **Programming Activity**	✓	✓		• PowerPoint presentation	• Seatwork • Quiz • Assignment • Class Participation • Major Exam

			values	• List Element Removal • List and Functions • List Methods • List Comprehensions • Multidimensional Lists • Lists vs Generators	• **Multimedia Presentation** • **Lecture** • **Code Demonstration**	✓				
			MIDTERM EXAMINATION							

LEARNING MATRIX

WEEK	HOUR		LEARNING OUTCOME	CONTENT	TEACHING-LEARNING ACTIVITY					ASSESSMENT
					Title	Modality **O**-Onsite, **A**-Offsite Asynchronous, **S**-Offsite Synchronous			Resource	
	Lec	Lab				**O**	**A**	**S**		
13th - 15th	2	3	At the end of the lesson, the learner will be able to: • Construct dictionaries and access entries in those dictionaries • Use methods to manipulate dictionaries • Decide whether a list or a dictionary is an appropriate data structure for a given application	**8. Tuples, Dictionaries and Sets** • Tuples • Dictionaries • Sets	• **Programming Activity**	✓	✓		• PowerPoint presentation	• Seatwork • Quiz • Assignment • Class Participation
					• **Multimedia Presentation** • **Lecture** • **Code Demonstration**	✓				
16th – 18th	2	3	• Open a text file for output and write strings or numbers to the file • Open a text for input and read strings or numbers from the file • Use library functions to access and navigate a file system	**9. Text Files** • Opening Files • Text Files and Lines • Reading Files • Searching through a file • Writing Files	• **Programming Activity**	✓	✓		• PowerPoint presentation	• Seatwork • Quiz • Assignment • Class Participation • Major Exam
					• **Multimedia Presentation** • **Lecture** • **Code Demonstration**	✓				
			FINAL EXAMINATION							